Linear Algebra : [ML] :

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Vector:

$[x_1, x_2, x_3, \ldots, x_n]$.

feature vectors:

we can behave input as feature columns:

[5.2, 2.1, 7.2, 9.1]

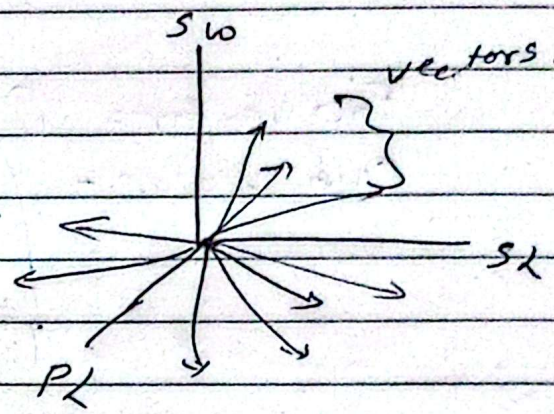
↑ 4d vector.

↓ ml model

↓ output.

SL	SW	PL	PW	targets ↓ species
5.2	2.1	2.3	2.3	Setosa
3.1	4.2	7	6.5	Variica

features.



- we send numbers in feature vectors not alphabets,

  if we have to send "words", then we should use "Encoding".

↓

NLP (BOW)

If we have to deal with:

sex
M → 0
F → 1.

Hello, everyone
It's

Hello	everyone	It	is
1	1	0	0
0	0	1	1

Like, ~~One Hot Encoding~~

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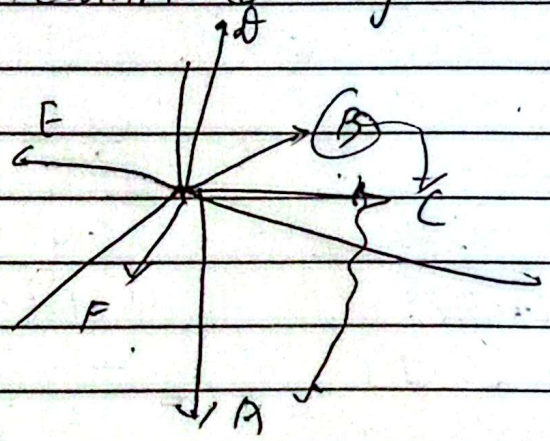
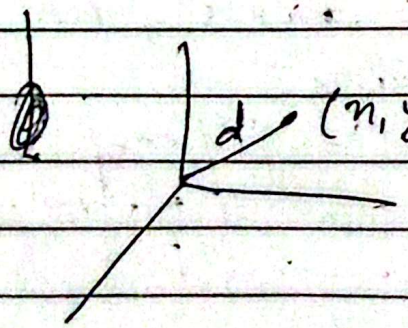
Movie Recommender system:



If you watch (B) then, it recommend C, D coz nearer in distance.

Row vector = $[x_1 \; - - - - \; x_n]_{1 \times n}$ ← colⁿ.

Colⁿ vector = $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$



$\Rightarrow d = \sqrt{x^2 + y^2 + z^2}$.

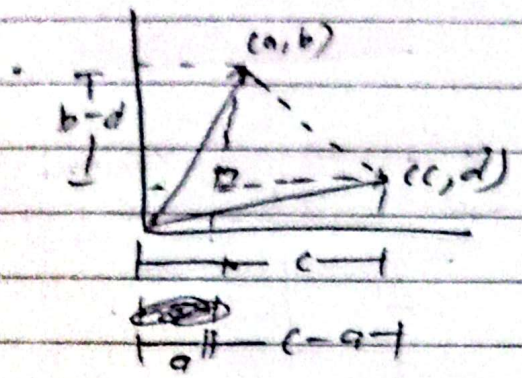
In n-D.

$d = \sqrt{x_1^2 + x_2^2 + x_3^2 + - - - x_n^2}$

magnitude: $\|A\|$

distance = np.linalg.norm(A)

Euclidean Distance.



2-D

$$dist = \sqrt{(c-a)^2 + (d-b)^2}$$

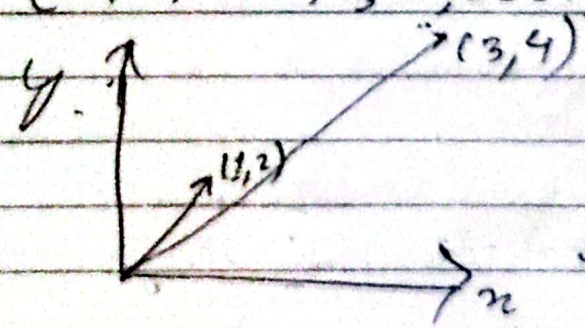
3-D.

$$distance = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2 + (z_1-z_2)^2}$$

diff = A - B.
distance = np.linalg.norm(diff)

~~Hider~~ Scalar Adding / Subtracting (Shifting).

$$V = [v_1, v_2, v_3, \ldots, v_n]$$
$$V + n = [v_1+n, v_2+n, v_3+n, \ldots, v_n+n]$$



scale = 2.

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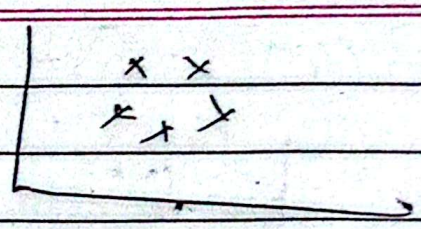
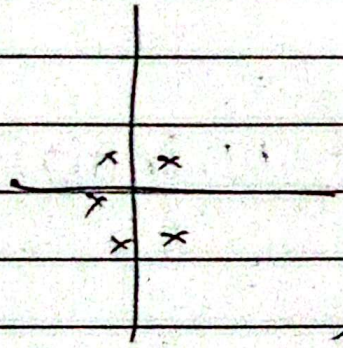
ml: scatter plot



	x_1	x_2
v_1	3	4
v_2	5	6
v_3	7	8.
mean:	5	6

x_1 shift.	x_2 shifted
-2	-2
0	0
2	2.



mean centering.

[PCA, Linear regression, clustering]

generate random value:

data = np.random.rand (100, 2)

100 vectors

2-d.

Inner product:

Dot Product: $\sum_{i=1}^{d} x_i y_i = x^T y = x \cdot y$

$x_1 y_1 + x_2 y_2 + \cdots + x_d y_d.$

~~Projection~~.

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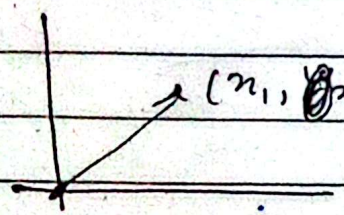
eq of plane:

$w_1 x_1 + w_2 x_2 + w_0 = 0. \rightarrow 2D$

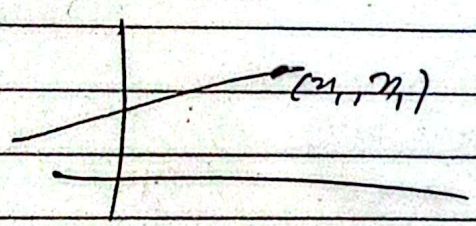
$3D \rightarrow w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 = 0.$

$nD \rightarrow w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w = 0.$



pass through

$(x_1, x_2) \rightarrow$ origin : $w_1 x_1 + w_2 x_2 = 0.$



$(x_1, x_2) \rightarrow w_1 x_1 + w_2 x_2 + w_0 = 0.$

$[w_1, w_2]$
$[x_1, x_2].$

$\boxed{w_1 x_1 + w_2 x_2}$
dot product.

$[w_1, w_2, w_3, w_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$